

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
- ✓ **DPP, REVISION SHEETS**
- ✓ **100% MONEYBACK TO MERITORIOUS STUDENTS**

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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Binomial Theorem

Q1

If sum of all the coefficient of even powers in $(1-x+x^2-x^3+\dots+x^{2n})(1+x+x^2+x^3+\dots+x^{2n})$ is 61 then n is equal to

$(1-x+x^2-x^3+\dots+x^{2n})(1+x+x^2+x^3+\dots+x^{2n})$ में समघातों के सभी गुणांकों का योगफल 61 है। तब n का मान बराबर है—

(1) 30

(2) 32

(3) 28

(4) 36

7th Jan Morning

Sol

(1)

Let माना $(1-x+x^2+\dots)(1+x+x^2+\dots) = a_0 + a_1x + a_2x^2 + \dots$

put $x = 1$ रखने पर

$$1(2n+1) = a_0 + a_1 + a_2 + \dots + a_{2n} \quad \dots\dots(i)$$

put $x = -1$ रखने पर

$$(2n+1) \times 1 = a_0 - a_1 + a_2 - \dots + a_{2n} \quad \dots\dots(ii)$$

Form (i) + (ii) से

$$4n + 2 = 2(a_0 + a_2 + \dots)$$

$$= 2 \times 61$$

$$\Rightarrow 2n+1 = 61 \Rightarrow n = 30$$

02

If $({}^{36}C_{r+1}) \times (k^2 - 3) = {}^{35}C_r \cdot 6$, then number of ordered pairs (r, k) are —(where $k \in I$).

यदि $({}^{36}C_{r+1}) \times (k^2 - 3) = {}^{35}C_r \cdot 6$, तो क्रमित युग्मों (r, k) की संख्या है —(जहाँ $k \in I$).

(1) 6

(2) 2

(3) 3

(4) 4

7th Jan Evening

Sol

(4)

$$\frac{36}{r+1} \times \frac{35}{C_r} (k^2 - 3) = \frac{35}{C_r}$$

$$k^2 - 3 = \frac{r+1}{6} \Rightarrow k^2 = 3 + \frac{r+1}{6}$$

r can be 5, 35

for $r = 5, k = \pm 2$ $r = 35, k = \pm 3$

Hence number of order pair = 4

03

Coefficient of x^7 in $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + \dots + x^{10}$ is- $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + \dots + x^{10}$ में x^7 का गुणांक है-

(1) 330

(2) 210

(3) 420

(4) 260

7th Jan Evening

Sol

(1)

$$(1+x)^{10} \left[1 - \left(\frac{x}{1+x} \right)^{11} \right]$$

$$\frac{(1+x)^{10} \left[(1+x)^{11} - x^{11} \right]}{(1+x)^{11} \times \frac{1}{(1+x)}}$$

$$(1+x)^{10} \left[(1+x)^{11} - x^{11} \right]$$

$$(1+x)^{11} \times \frac{1}{(1+x)}$$

$$= (1+x)^{11} - x^{11}$$

$$\text{Coefficient of } x^7 \text{ is } {}^{11}C_7 = {}^{11}C_4 = 330$$

$$x^7 \text{ का गुणांक } {}^{11}C_7 = {}^{11}C_4 = 330$$

04

If maximum value of ${}^{19}C_p$ is a, ${}^{20}C_q$ is b, ${}^{21}C_r$ is c, then relation between a, b, c isयदि ${}^{19}C_p$ का अधिकतम मान a, ${}^{20}C_q$ का अधिकतम मान b तथा ${}^{21}C_r$ का अधिकतम मान c है, तब a, b, c में सम्बन्ध है

$$(1) \frac{a}{11} = \frac{b}{22} = \frac{c}{42} \quad (2) \frac{a}{22} = \frac{b}{11} = \frac{c}{42} \quad (3) \frac{a}{22} = \frac{b}{42} = \frac{c}{11} \quad (4) \frac{a}{21} = \frac{b}{11} = \frac{c}{22}$$

8th Jan Morning

Sol

(1)

We know nC_r is max at middle termहम जानते हैं कि nC_r मध्य पद पर अधिकतम होता है।

$$a = {}^{19}C_p = {}^{19}C_{10} = {}^{19}C_9$$

$$b = {}^{20}C_q = {}^{20}C_{10}$$

$$c = {}^{21}C_6 = {}^{21}C_{10} = {}^{21}C_{11}$$

$$\frac{a}{{}^{19}C_9} = \frac{b}{{}^{20}C_{10}} = \frac{c}{{}^{21}C_{11}}$$

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{42/11}$$

$$\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$$

05

Let coefficient of x^4 and x^2 in the expansion of $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$ is α and β then $\alpha - \beta$ is equal to

माना $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$ के प्रसार में x^4 तथा x^2 के गुणांक क्रमशः α तथा β हैं तो $\alpha - \beta$ का मान है :

(1) 48

(2) 60

(3) -132

(4) -60

8th Jan Evening

Sol

(3)

$$\begin{aligned} & 2[{}^6C_0 x^6 + {}^6C_2 x^4 (x^2 - 1) + {}^6C_4 x^2 (x^2 - 1)^2 + {}^6C_6 (x^2 - 1)^3] \\ &= 2[x^6 + 15(x^6 - x^4) + 15x^2(x^4 - 2x^2 + 1) + (-1 + 3x^2 - 3x^4 + x^6)] \\ &= 2(32x^6 - 48x^4 + 18x^2 - 1) \end{aligned}$$

$$\alpha = -96 \text{ and } \beta = 36 \therefore \alpha - \beta = -132$$

06

Find the coefficient of x^4 in $(1 + x + x^2)^{10}$ 9th Jan Morning

Sol

615

General term महत्तम पद $\frac{10!}{\alpha!\beta!\gamma!} x^{\beta+2\gamma}$

for coefficient of x^4 का गुणांक $\Rightarrow \beta + 2\gamma = 4$

$$\gamma = 0, \beta = 4, \alpha = 6 \Rightarrow \frac{10!}{6!4!0!} = 210$$

$$\gamma = 1, \beta = 2, \alpha = 7 \Rightarrow \frac{10!}{7!2!1!} = 360$$

$$\gamma = 2, \beta = 0, \alpha = 8 \Rightarrow \frac{10!}{8!0!2!} = 45$$

Total कुल = 615

07

If ${}^{25}C_0 + 5 {}^{25}C_1 + 9 {}^{25}C_2 + \dots + 101 {}^{25}C_{25} = 2^{25}k$ find $k = ?$

9th Jan Evening

Sol

(51)

$$\begin{aligned} \sum_{r=0}^{25} (4r+1) {}^{25}C_r &= 4 \sum_{r=0}^{25} r \cdot {}^{25}C_r + \sum_{r=0}^{25} {}^{25}C_r \\ &= 4 \sum_{r=1}^{25} r \times \frac{25}{r} {}^{24}C_{r-1} + 2^{25} = 100 \sum_{r=1}^{25} {}^{24}C_{r-1} + 2^{25} \\ &= 100 \cdot 2^{24} + 2^{25} = 2^{25}(50 + 1) = 51 \cdot 2^{25} \end{aligned}$$

So इसलिये $k = 51$

08

If minimum value of term free from x for $\left(\frac{x}{\sin \theta} + \frac{1}{x \cos \theta} \right)^{16}$ is L_1 in $\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4} \right]$ and L_2 in $\theta \in \left[\frac{\pi}{16}, \frac{\pi}{8} \right]$ find

$$\frac{L_2}{L_1}$$

9th Jan Evening

Sol

16

$$T_{r+1} = {}^{16}C_r \left(\frac{x}{\sin \theta} \right)^{16-r} \left(\frac{1}{x \cos \theta} \right)^r$$

for $r = 8$ term is free from 'x'

$$T_9 = {}^{16}C_8 \frac{1}{\sin^8 \theta \cos^8 \theta}$$

$$T_9 = {}^{16}C_8 \frac{2^8}{(\sin 2\theta)^8}$$

$$\text{in } \theta \in \left[\frac{\pi}{8}, \frac{\pi}{4} \right], L_1 = {}^{16}C_8 2^8 \quad \because \{ \text{Min value of } L_1 \text{ at } \theta = \pi/4 \}$$

$$\text{in } \theta \in \left[\frac{\pi}{16}, \frac{\pi}{8} \right], L_2 = {}^{16}C_8 \frac{2^8}{\left(\frac{1}{\sqrt{2}} \right)^8} = {}^{16}C_8 \cdot 2^8 \cdot 2^4 \quad \{ \because \text{min value of } L_2 \text{ at } \theta = \pi/8 \}$$

$$\frac{L_2}{L_1} = \frac{{}^{16}C_8 \cdot 2^8 \cdot 2^4}{{}^{16}C_8 \cdot 2^8} = 16$$



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